

FPGAs IN DATA CENTRES: Opportunities And Challenges



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Catapult cloud field-programmable gate array (FPGA) architecture accelerates both cloud services (such as search engine) and the Azure cloud platform; Azure is an open, flexible, enterprise-grade cloud computing platform. The reconfigurable Catapult fabric is embedded into each half-rack of 48 servers in the form of a small board with a medium-sized FPGA and local dynamic random-access memory (DRAM) attached to each server. FPGAs used in Catapult servers are central to delivering better Bing results. These can quickly score, filter, rank and measure the relevancy of text and image queries on Bing.

Catapult v2 design is more flexible in circumventing traditional data centre structures for machine learning and expands the role of FPGAs as accelerators. It expands the availability of FPGAs, allowing them to be hooked up to a larger number of computing resources. FPGAs are connected to DRAM, central processing unit (CPU) and network switches. These can accelerate local applications, or act as a processing resource in large-scale, deep-learning models. Much like with Bing, FPGAs can be involved in scoring results and training deep-learning models.

Catapult v2 design can be used for cloud-based image recognition, natural language processing and other tasks typically associated with machine learning. It could also provide a blueprint for using FPGAs in machine learning installations. Many ma-

chine learning models are driven by graphical processing units (GPUs), but the role of FPGAs is less clear. FPGAs can quickly deliver deep-learning results, but consume too much power if not programmed correctly. These can be reprogrammed to execute specific tasks, but that also makes them one-dimensional. In comparison, GPUs are more flexible and capable of handling several calculations.

Catapult server specification

Project Catapult employs an elastic architecture that links FPGAs together in a 6x8

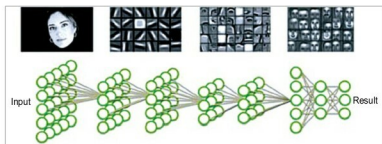


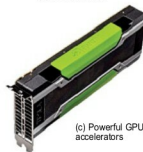
Fig. 16: Deep learning



(a) More data



(b) Better models



(c) Powerful GPU accelerators

Fig. 17: Drivers for deep learning